

Act IV: The Play Begins

Market Equilibrium, Price Setting, Taxes

Module XI: Town Crisis

Externalities and Public Goods



Previously in Marginal Falls

- ◆ Buyers gave us Demand.
- ◆ Sellers gave us Supply.
- ◆ The market found equilibrium.
- ◆ Fair Play measured surplus.

Next: What if the market misses someone?

The Smoke Complaint

Can a market be wrong even when buyers and sellers trade voluntarily?



The buyer and seller both agreed. So where is the problem?

Externalities

Externality:

A benefit or cost that affects someone who is not directly involved in the production or consumption of a good or service.

If pollution only affected the person who created it



Create pollution only until it's $MB > MC$

Types of Externalities

Negative Externality:

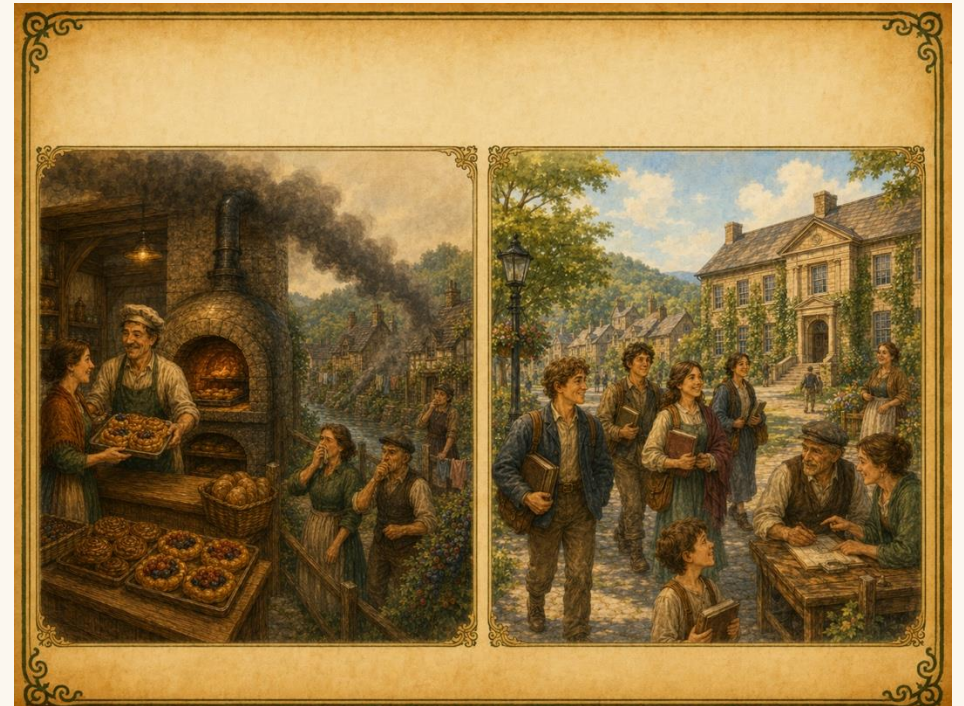
Social Benefit < Private Benefit

e.g. Smoking

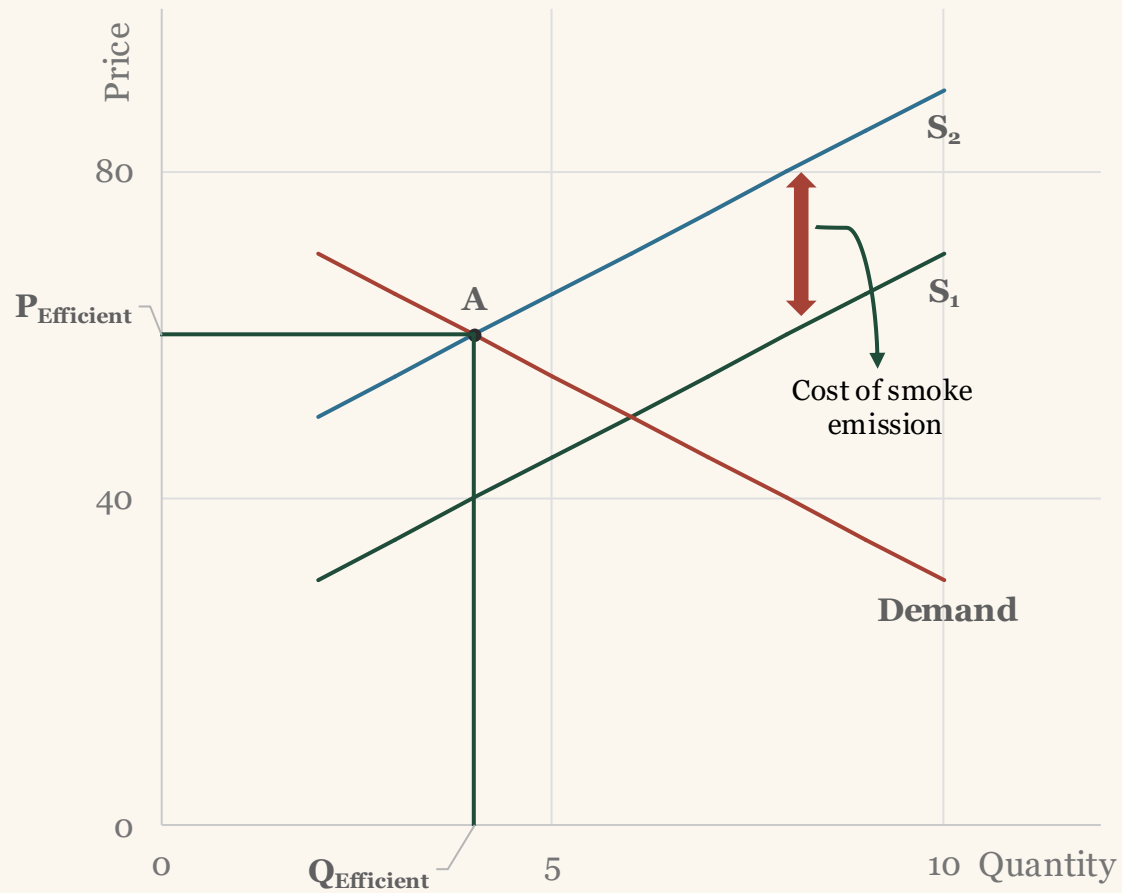
Positive Externality:

Social Benefit > Private Benefit

e.g. Education



Negative Externality

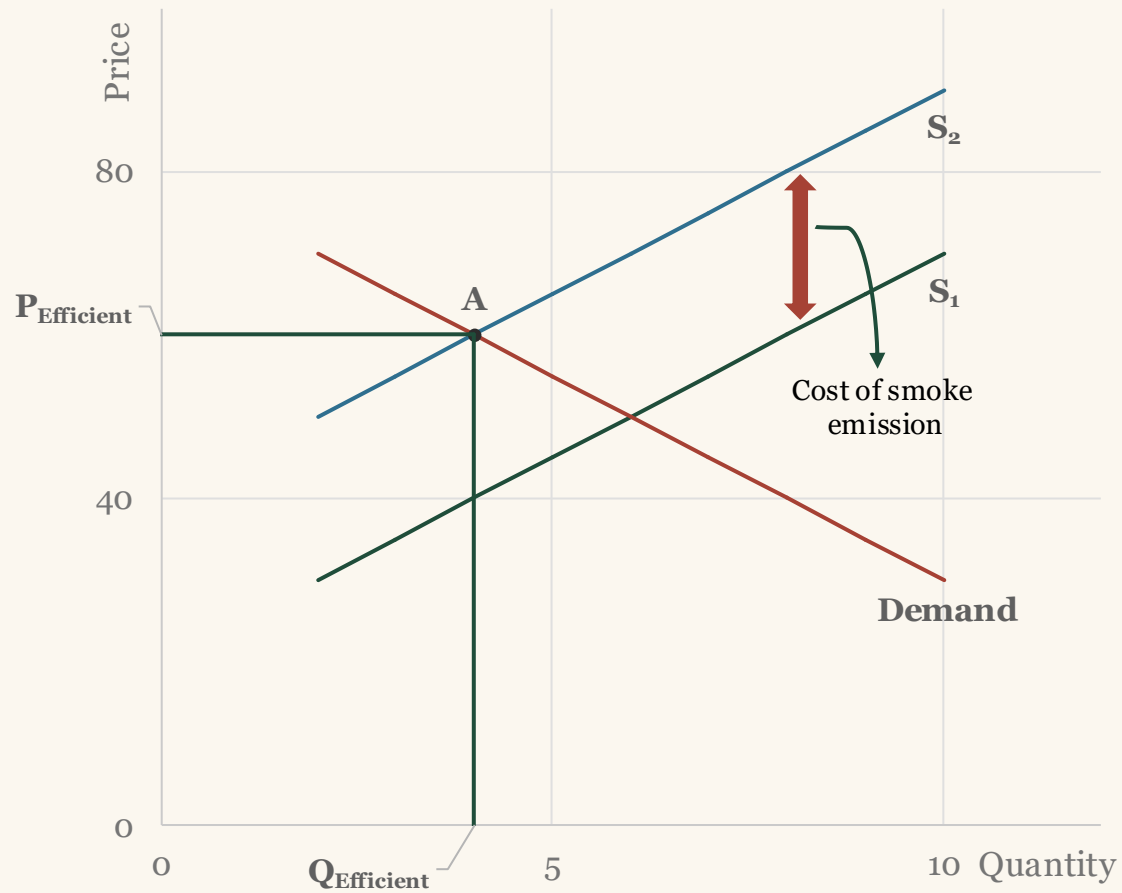


Private Cost:
what the bakery pays.

External Cost:
smoke damage to others.

**Social Cost = Private Cost +
External Cost**

Negative Externality

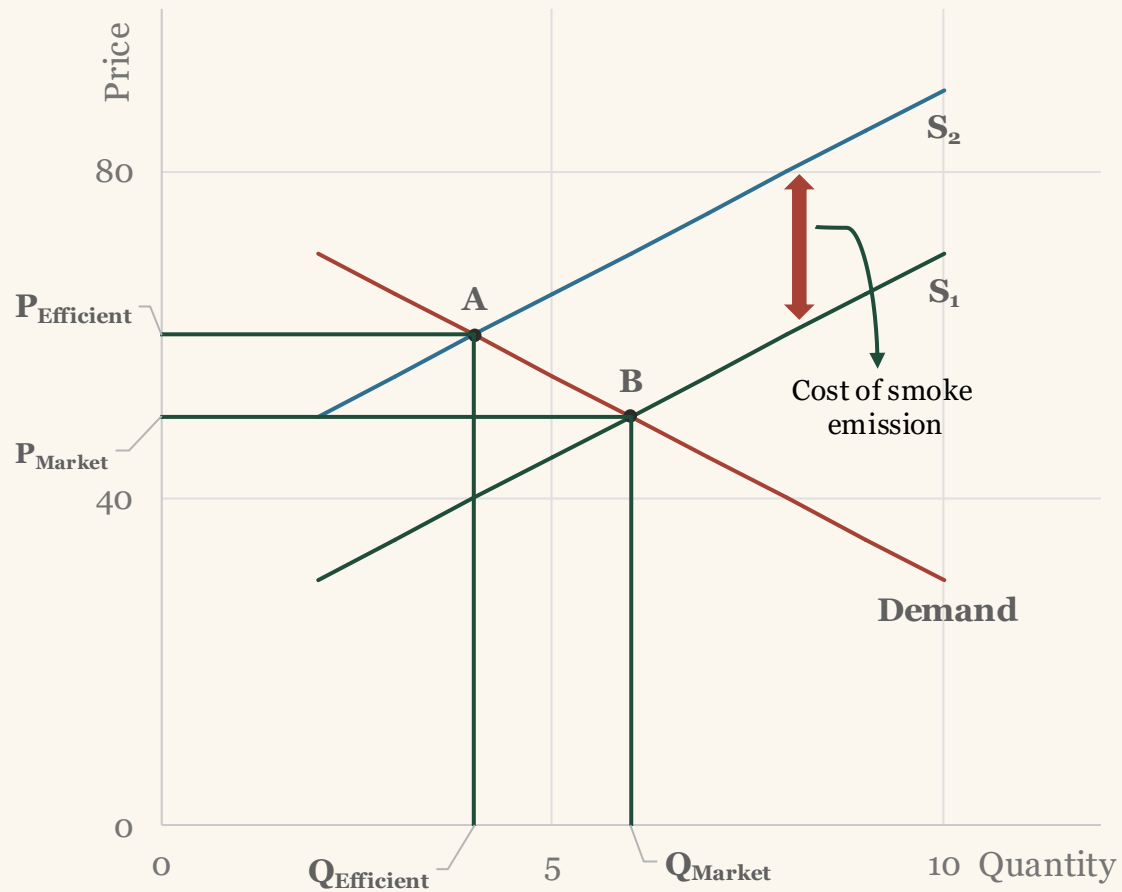


S_1 : Marginal Private Cost

S_2 : Marginal Social Cost

Efficient: $MB = MSC$

Negative Externality



S_1 : Marginal Private Cost

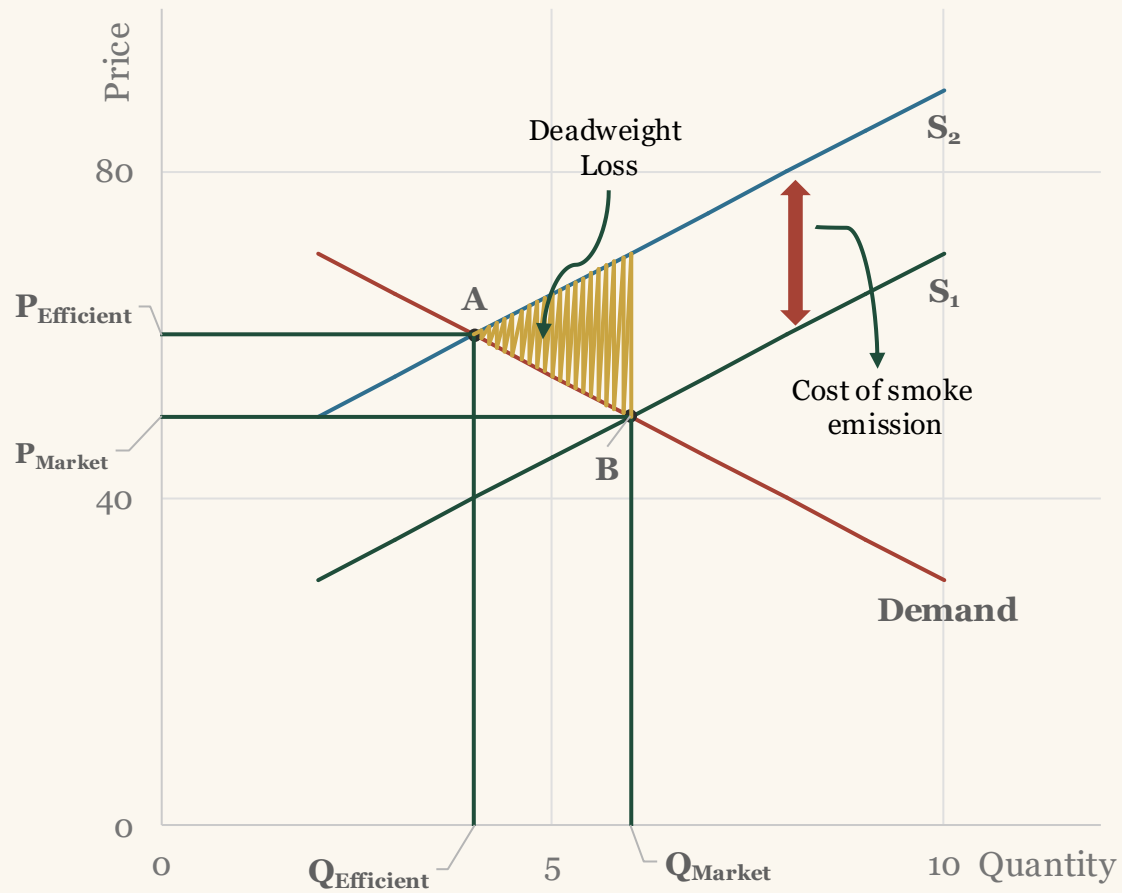
S_2 : Marginal Social Cost

Market: $MB = MPC$

Efficient: $MB = MSC$

The price P_{Market} is too low because smoke is not priced.

Negative Externality



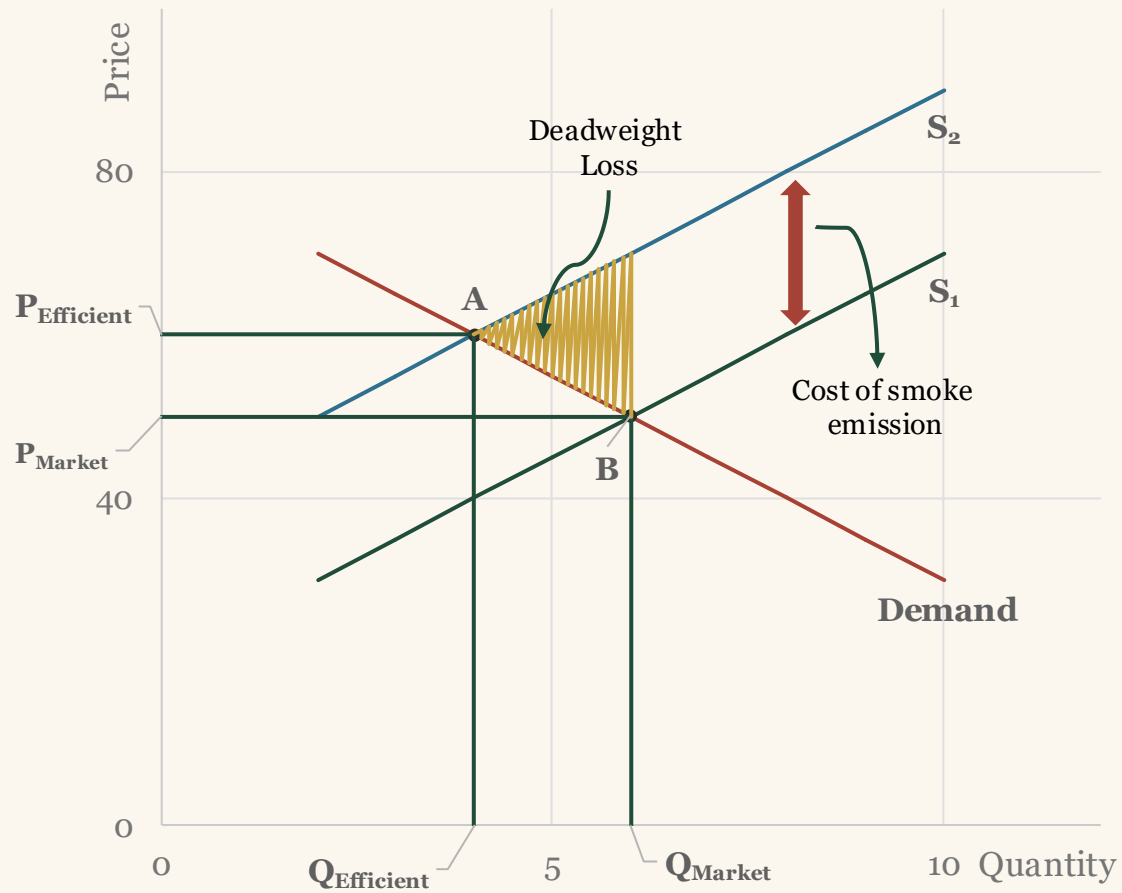
Market: $MB = MPC$

Efficient: $MB = MSC$

The price P_{Market} is too low because smoke is not priced.

Deadweight Loss is generated.

Negative Externality



General Rule:

Negative Externality: $Q_{\text{Market}} > Q_{\text{Efficient}}$

Too much of the good is produced.

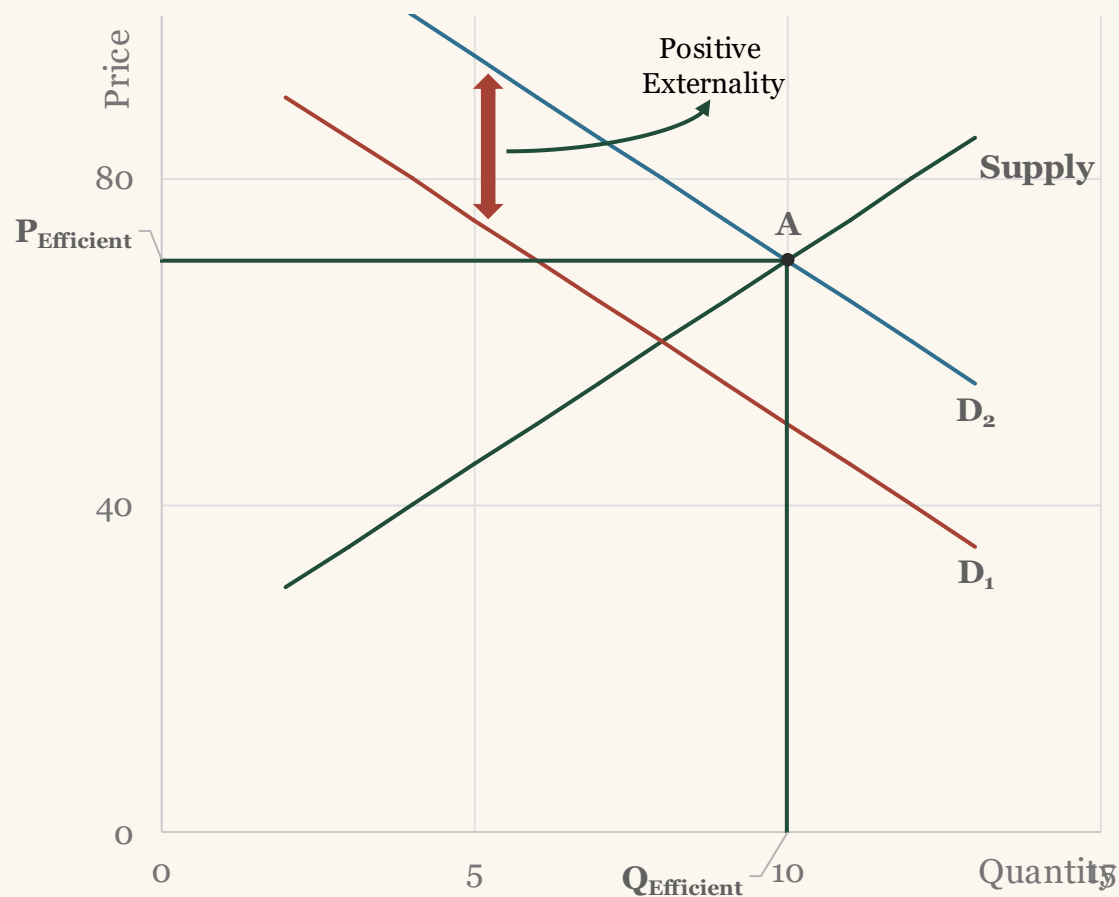
Positive Externality

**The Marginal Falls Academy
opens.**

A student gains:
better skills, better jobs, higher
income.

The town gains:
better decisions, new ideas,
stronger civic life.

Positive Externality



Private Benefit:

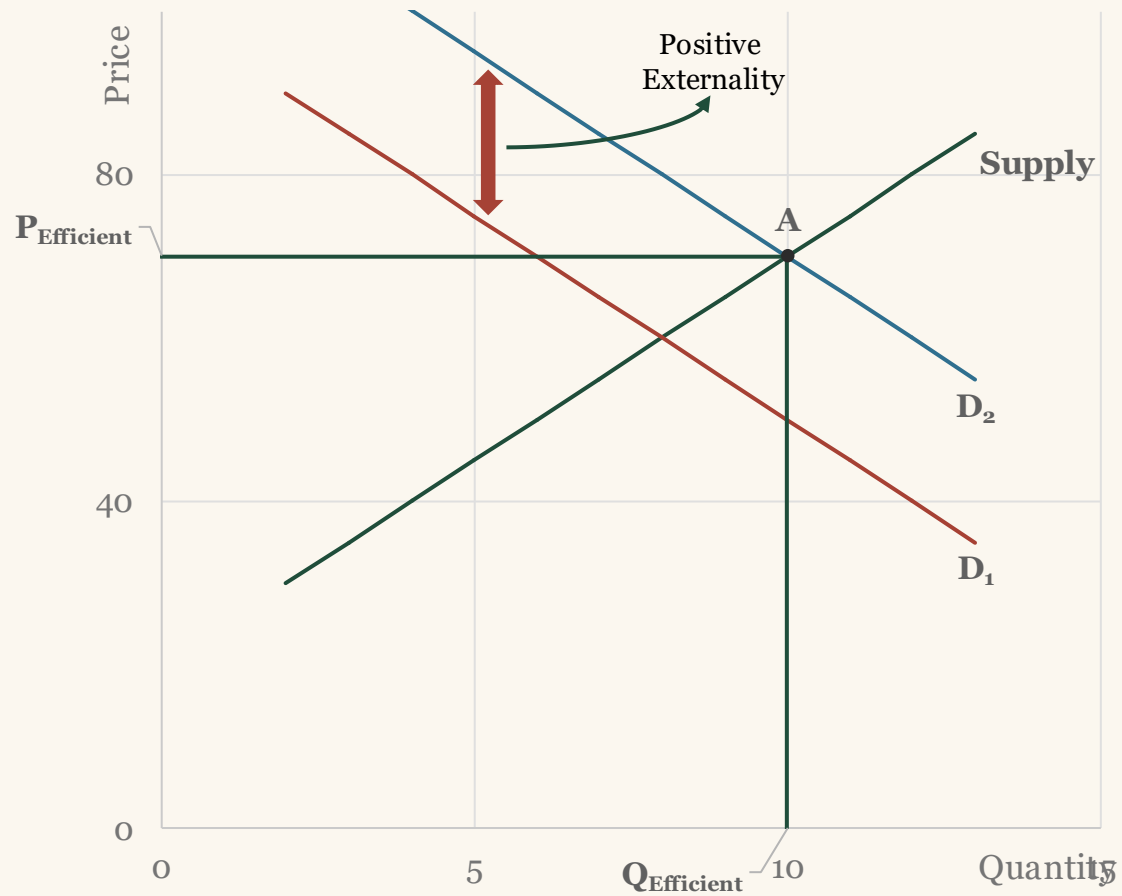
what each student gains from education.

External Benefit:

what others gain from educated people.

Social Benefit = Private Benefit + External Benefit

Positive Externality

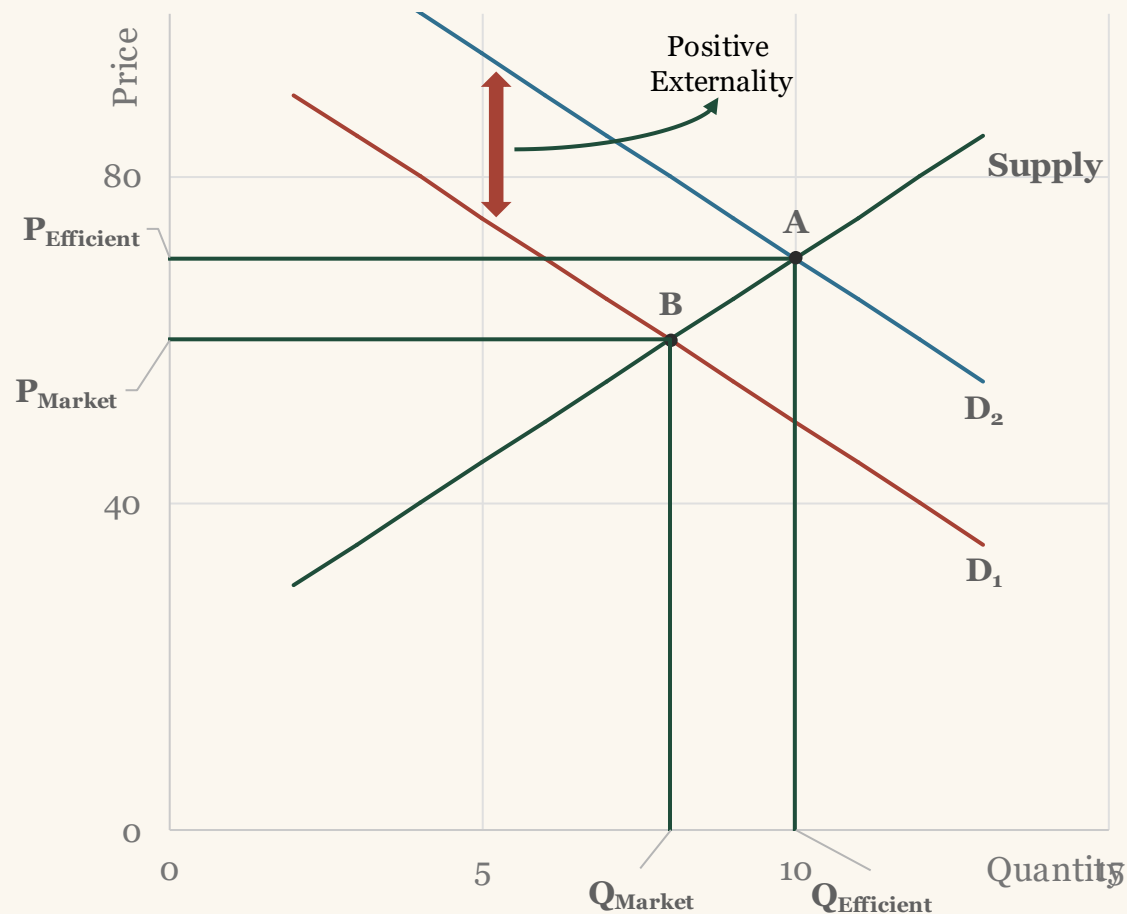


D₁: Marginal Private Benefit

D₂: Marginal Social Benefit

Efficient: $MSB=MC$

Positive Externality



D₁: Marginal Private Benefit

D₂: Marginal Social Benefit

Market: $MPB = MC$

Efficient: $MSB = MC$

The quantity of education Q_{Market} is too low because market only cares about private benefit.

Positive Externality



Market: $MPB = MC$

Efficient: $MSB = MC$

The quantity of education Q_{Market} is too low because market only cares about private benefit.

Deadweight Loss is generated.

Positive Externality

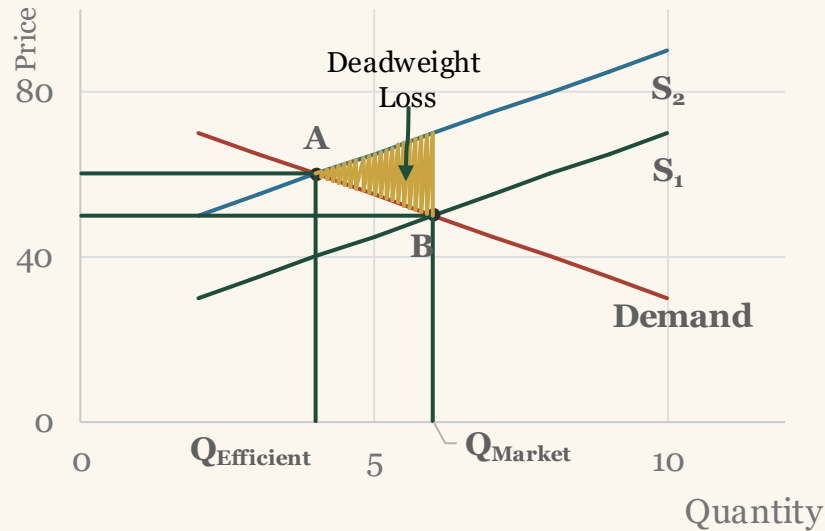


General Rule:

Positive Externality: $Q_{\text{Market}} < Q_{\text{Efficient}}$

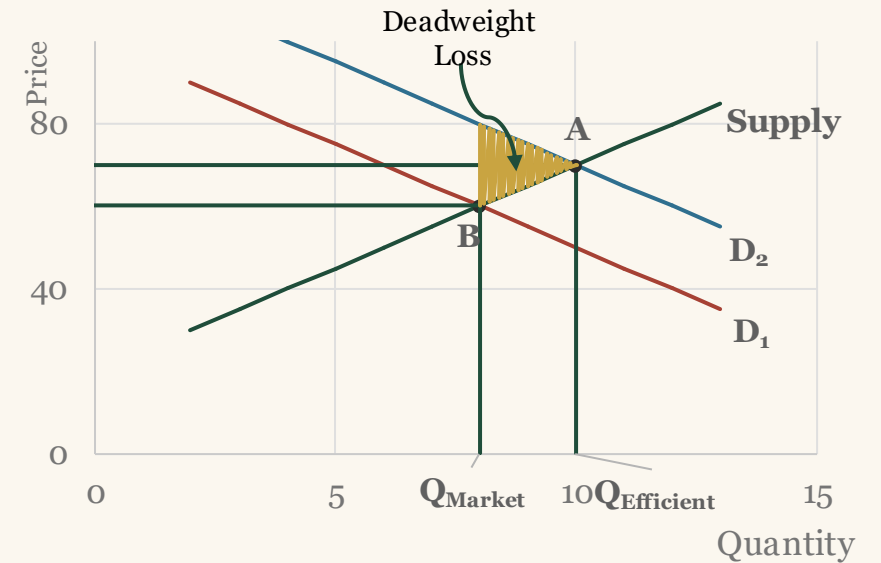
Too little of the good is produced.

Market Failure



Negative Externality

$$Q_{\text{Market}} > Q_{\text{Efficient}}$$



Positive Externality

$$Q_{\text{Market}} < Q_{\text{Efficient}}$$

Market Failure:

A situation in which the market fails to produce the efficient level of output.

Property Rights

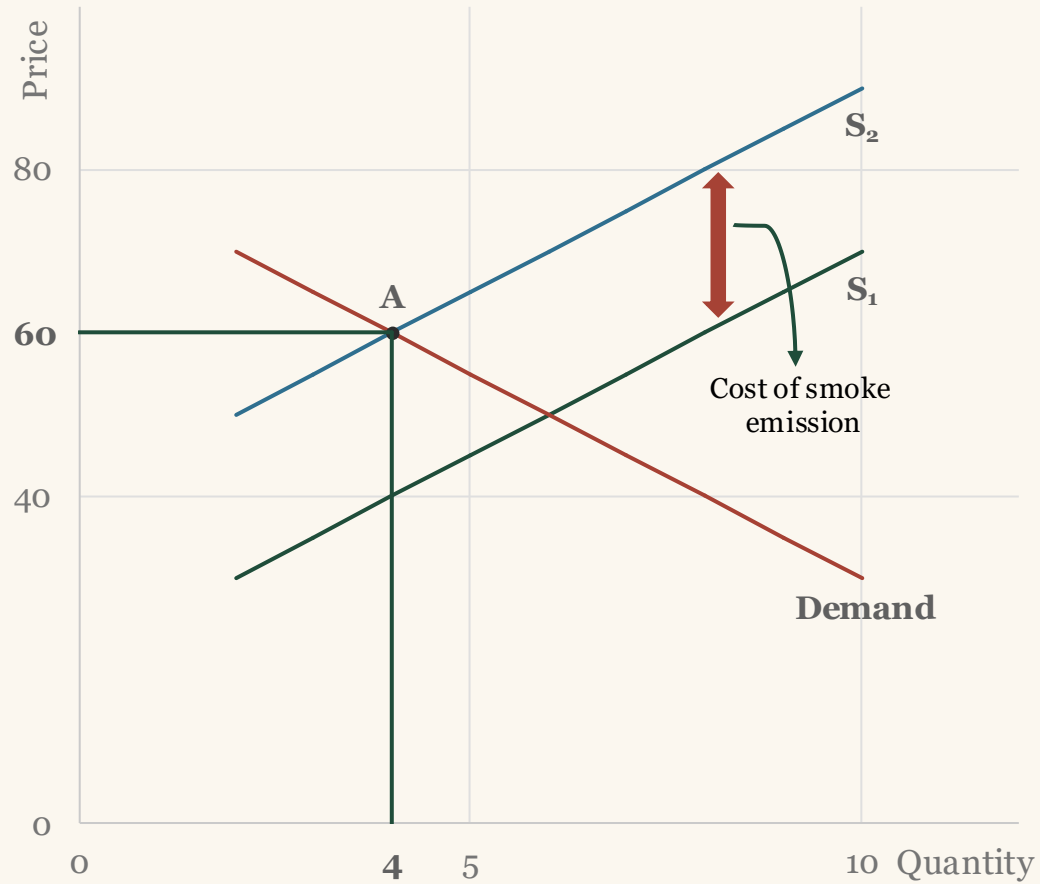
Property rights:

The rights individuals or businesses have to the exclusive use of their property, including the right to buy or sell it.

Marginal Falls:

- Bakery's right to release soot?
- Citizens' right to clean river water?

Back at the Bakery

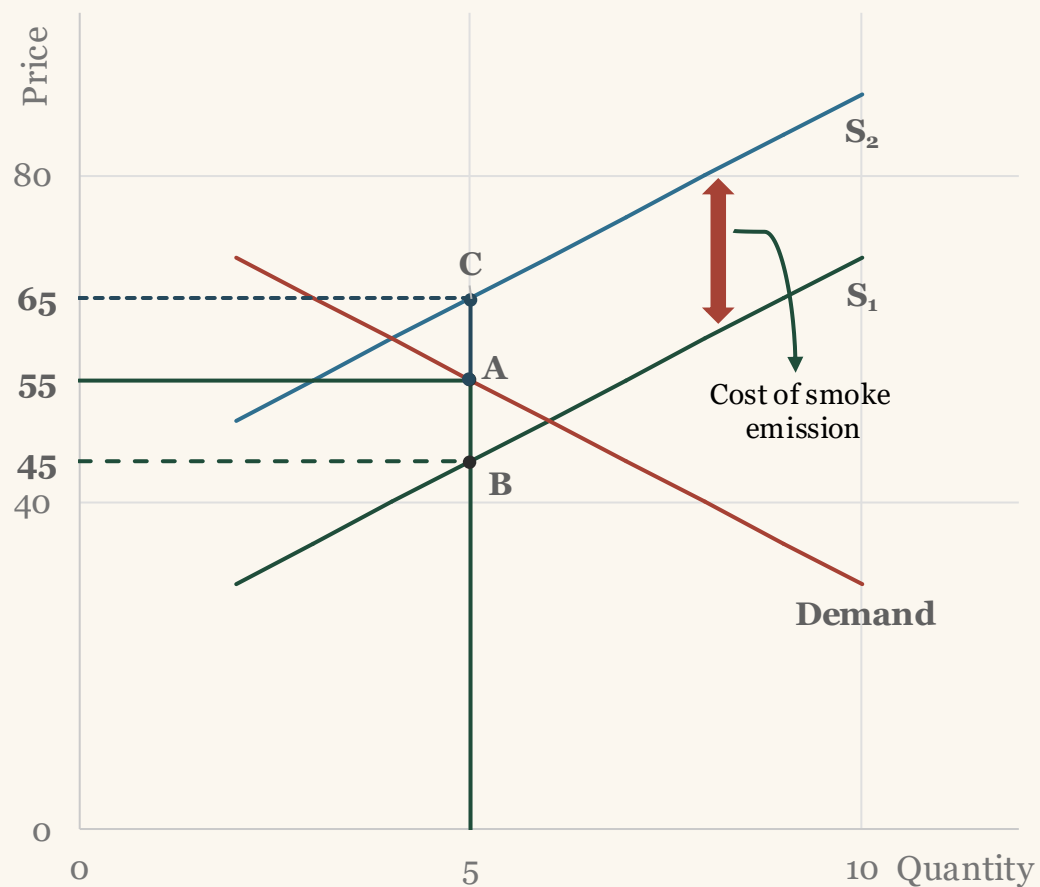


Efficient Outcome:

$Q = 4$ units

$P = 60$ Tokens

Back at the Bakery



Producing one more batch of Pastries:

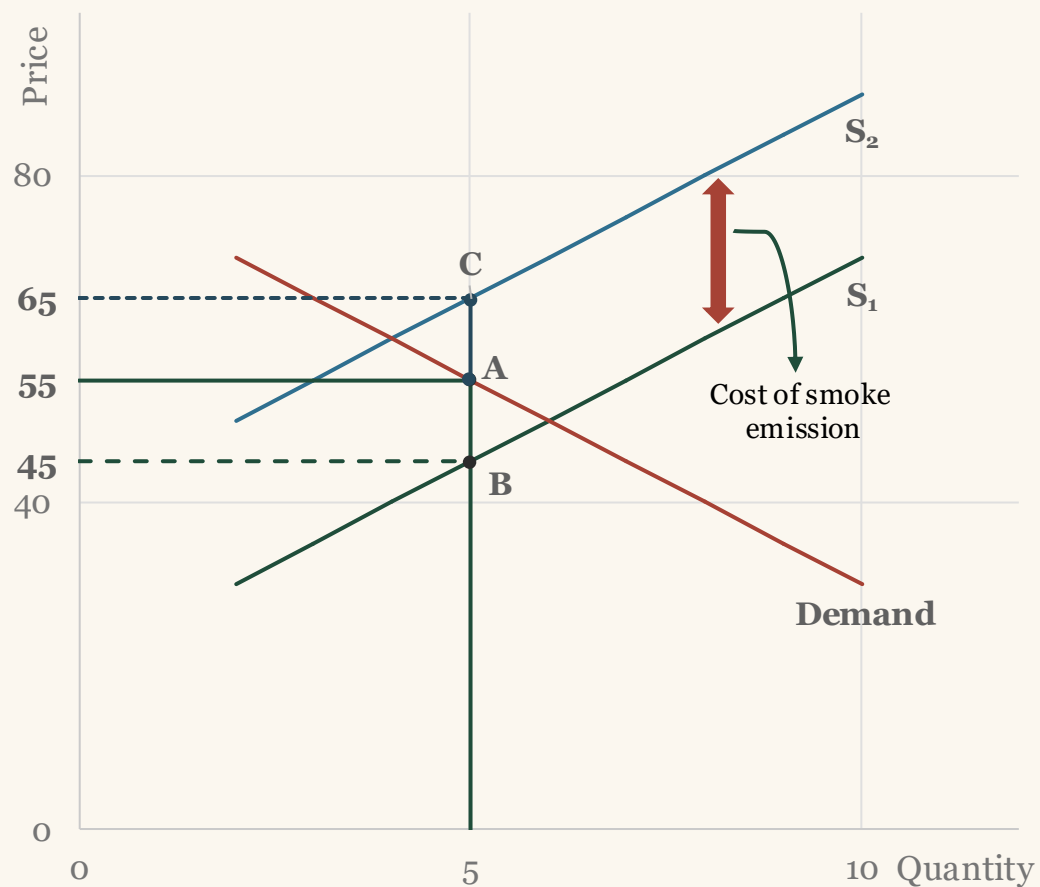
$$Q = 5$$

MB = 55 (point A)

MSC = 65 (point C)

MPC = 45 (point B)

Back at the Bakery



Producing one more batch of Pastries:

$$Q = 5$$

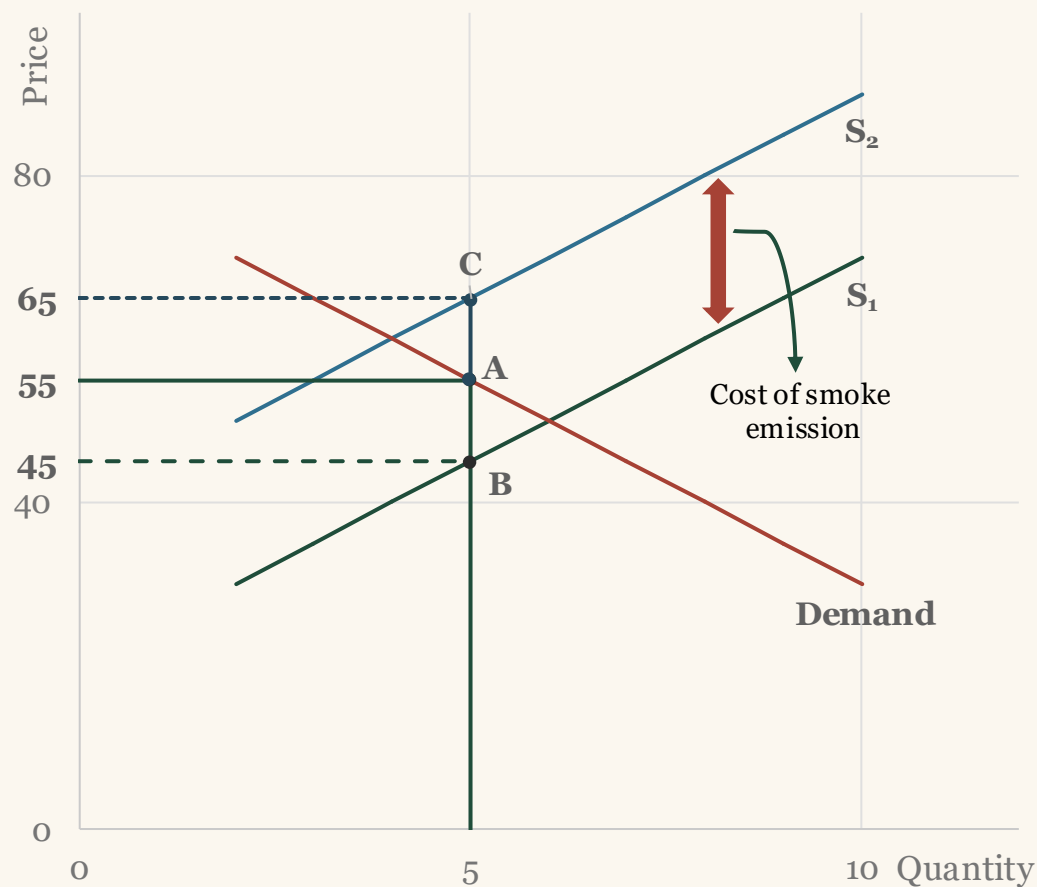
$$MB = 55$$

$$MSC = 65$$

$$MPC = 45$$

$$\text{External Cost} = MSC - MPC = 20$$

Back at the Bakery



Producing one more batch of
Pastries: $Q = 5$

$$\text{External Cost} = \text{MSC} - \text{MPC} = 20$$

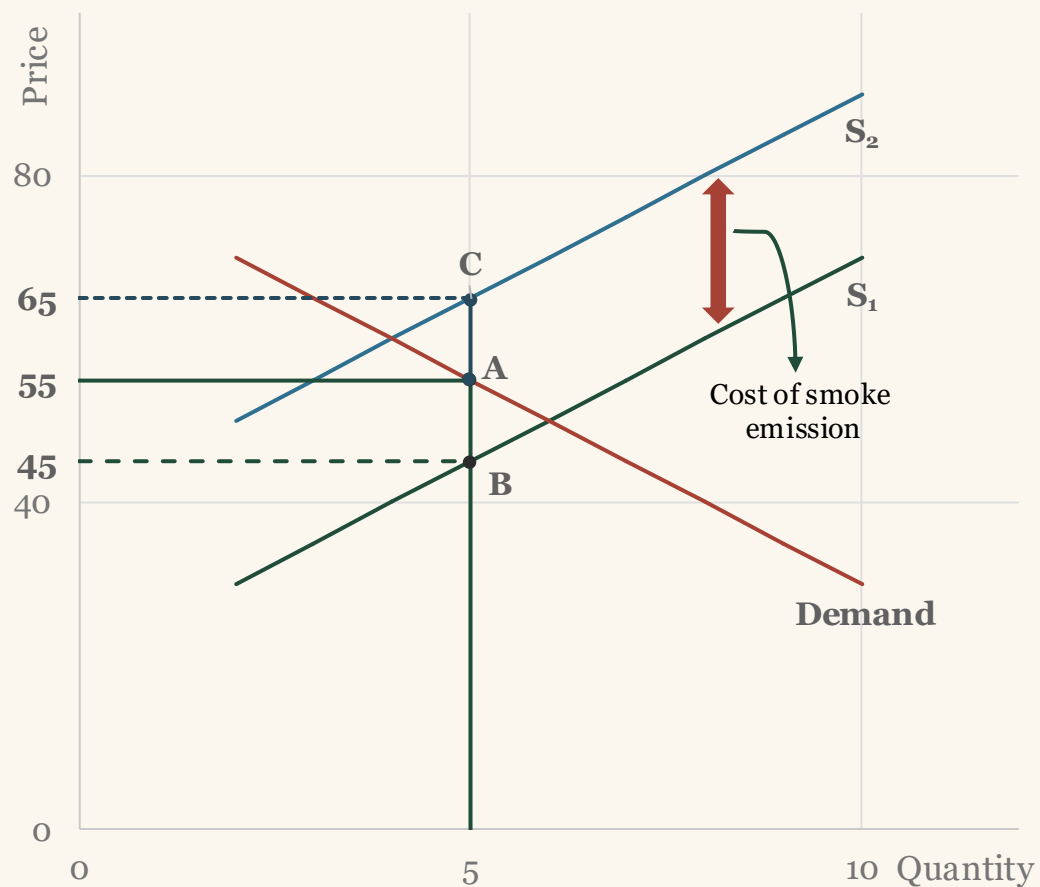
Private Gain:

$$\text{MB} - \text{MPC} = 55 - 45 = 10$$

Social Cost (Water Damage):

$$\text{External Cost} = 20$$

Back at the Bakery



Producing one more batch of Pastries: $Q = 5$

Private Gain:

$$MB - MPC = 55 - 45 = 10$$

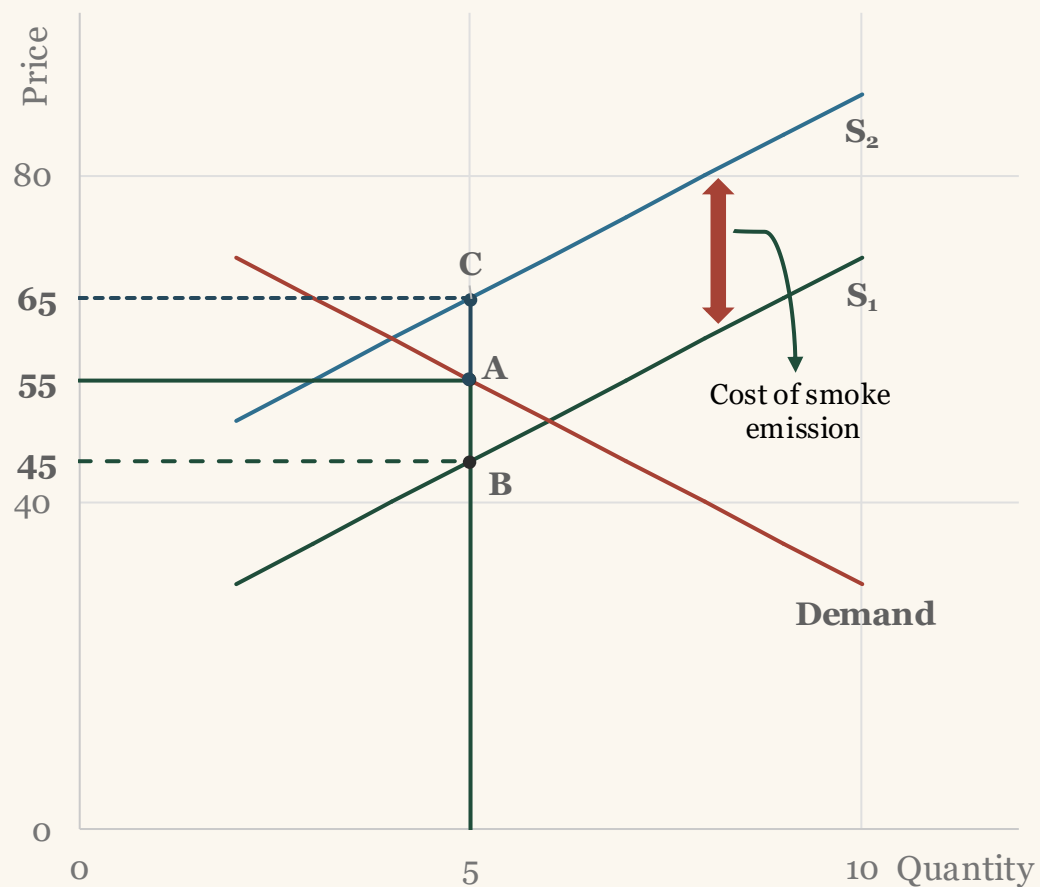
Social Cost (Water Damage):

$$\text{External Cost} = 20$$

Town Effect:

$$\begin{aligned} \text{Private Gain} - \text{Social Cost} \\ = 10 - 20 = -10 \end{aligned}$$

Back at the Bakery



Producing one more batch of
Pastries: $Q = 5$

Town Effect:

$$\begin{aligned} &\text{Private Gain} - \text{Social Cost} \\ &= 10 - 20 = -10 \end{aligned}$$

5th batch of pastry shouldn't be
produced.

Clean River Rights

Citizens have the *Clean River Right*

Bakery must buy permission.

- Bakery side will pay at most: 10
- Citizens require at least: 20



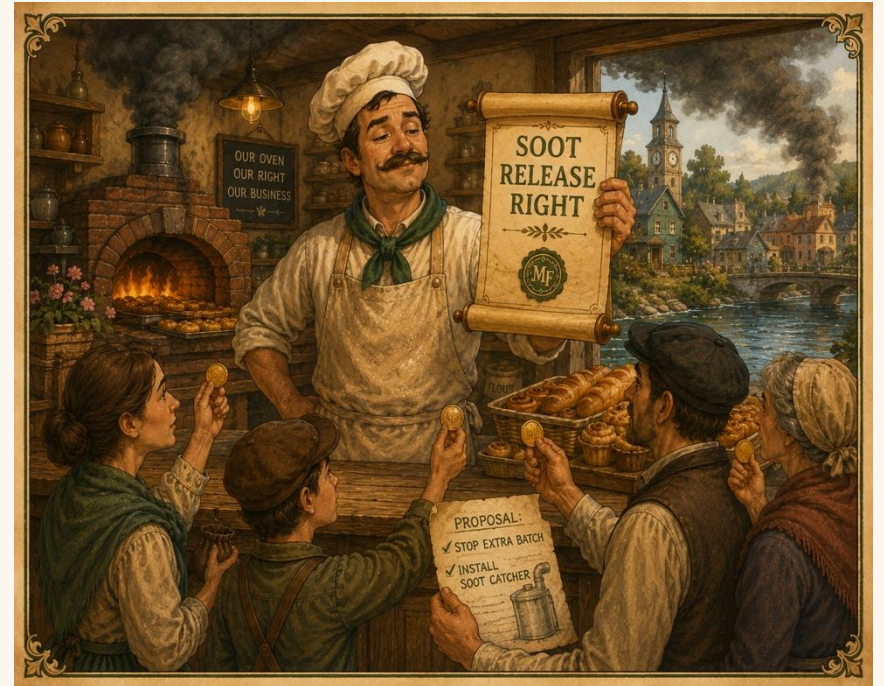
No Deal. No 5th batch of Pastries.

Soot Release Rights

Bakery has the *Soot Release Right*

Citizens must pay bakery to stop.

- Citizens will pay up to: 20
- Bakery side accepts at least: 10



Deal happens. No 5th batch.

Same Outcome, Different Payments

Efficient outcome: **No 5th batch.**

If citizens have the right:
Bakery does not buy permission.

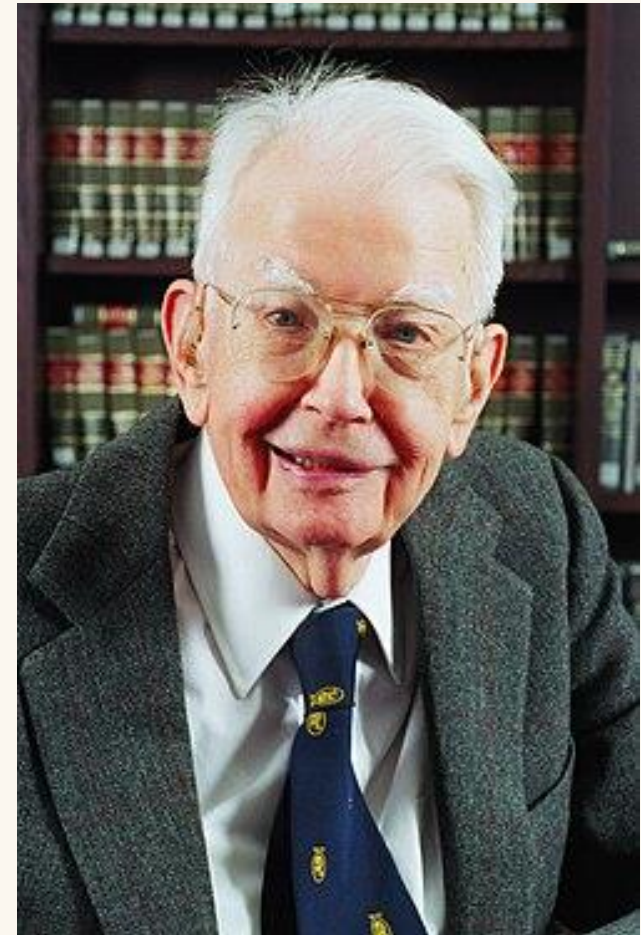
If bakery has the right:
Citizens pay bakery to stop.

Rights affect who pays. Efficiency can be the same.

Coase Theorem

Ronald Coase studied how property rights and transaction costs shape markets.

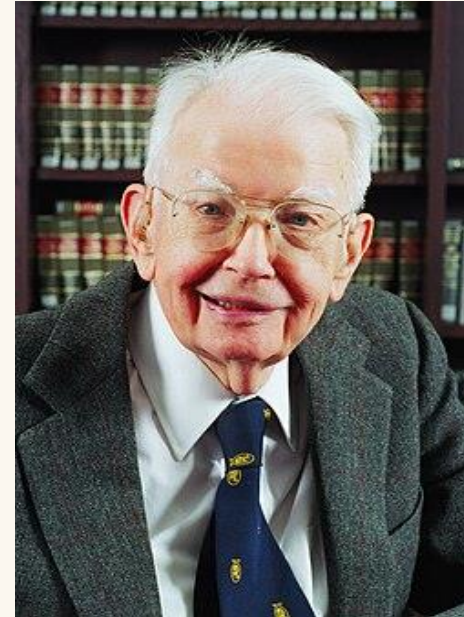
He won the 1991 Nobel Prize in Economics.



Coase Theorem

Private bargaining can solve externalities if:

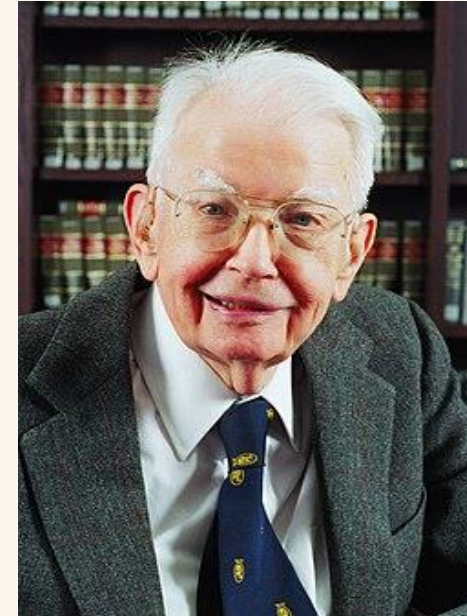
- Property rights are clear.
- Transaction costs are low.
- Parties have full information.



Coase Theorem

Private bargaining can solve externalities if:

- Property rights are clear.
- Transaction costs are low.
- Parties have full information.



Transactions costs:

The costs in time and other resources that parties incur in the process of agreeing to and carrying out an exchange of goods or services.

The Mayor's Toolkit

To fix externalities, the Mayor can use:

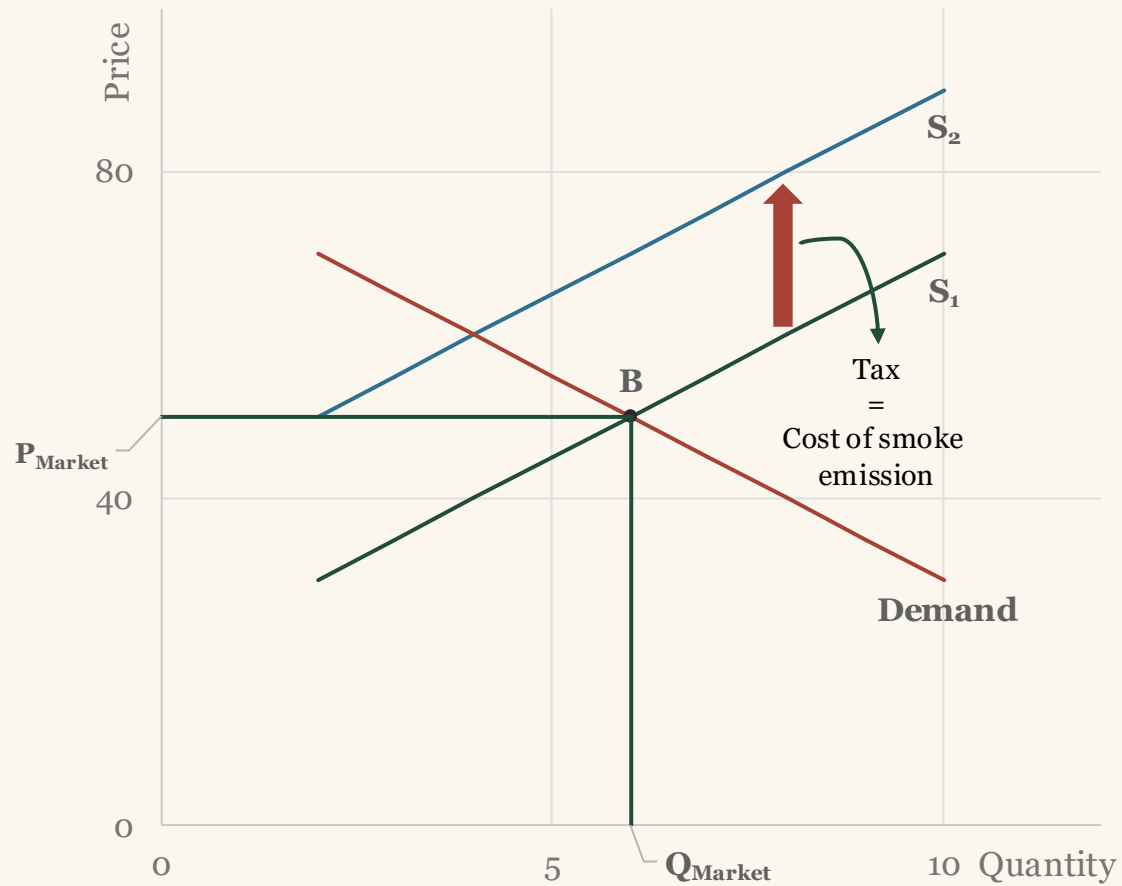
Taxes
Subsidies
Rules
Permits



Goal:

Make private choices reflect social costs and benefits.

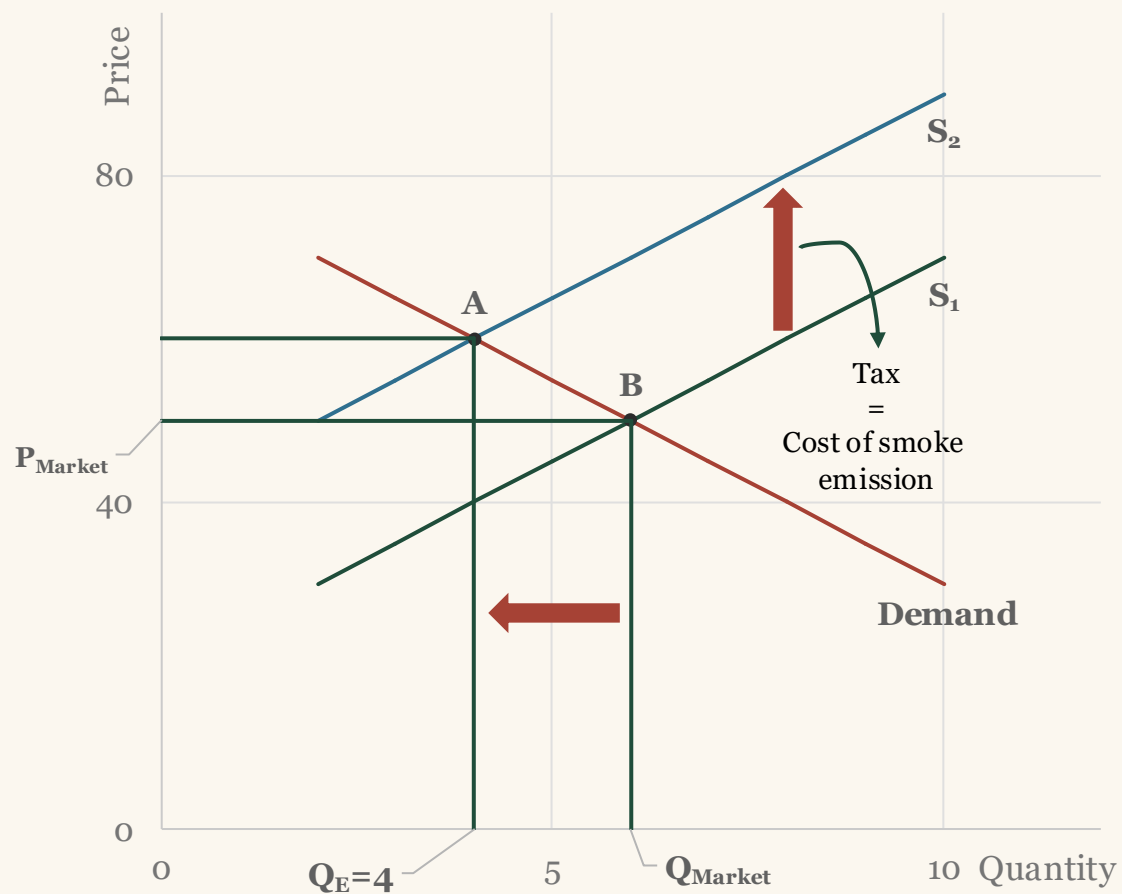
Corrective Taxes



Tax = Cost of Smoke Emission

Tax = 20 Tokens per batch sold

Corrective Taxes



Tax = Cost of Smoke Emission

Tax = 20 Tokens per batch sold

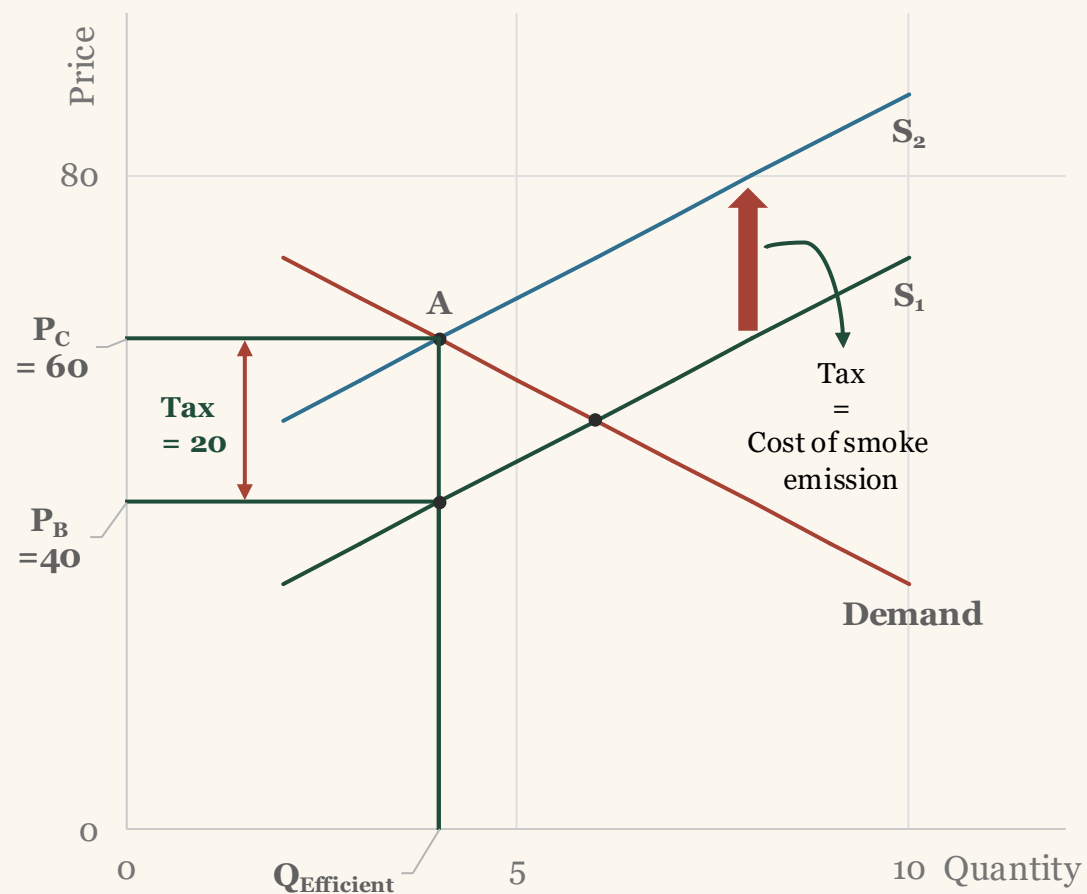
Before Tax: $MB = MPC$

$$Q_M = 6$$

After Tax: $MB = MSC$

$$Q_E = 4$$

Corrective Taxes



Tax = Cost of Smoke Emission

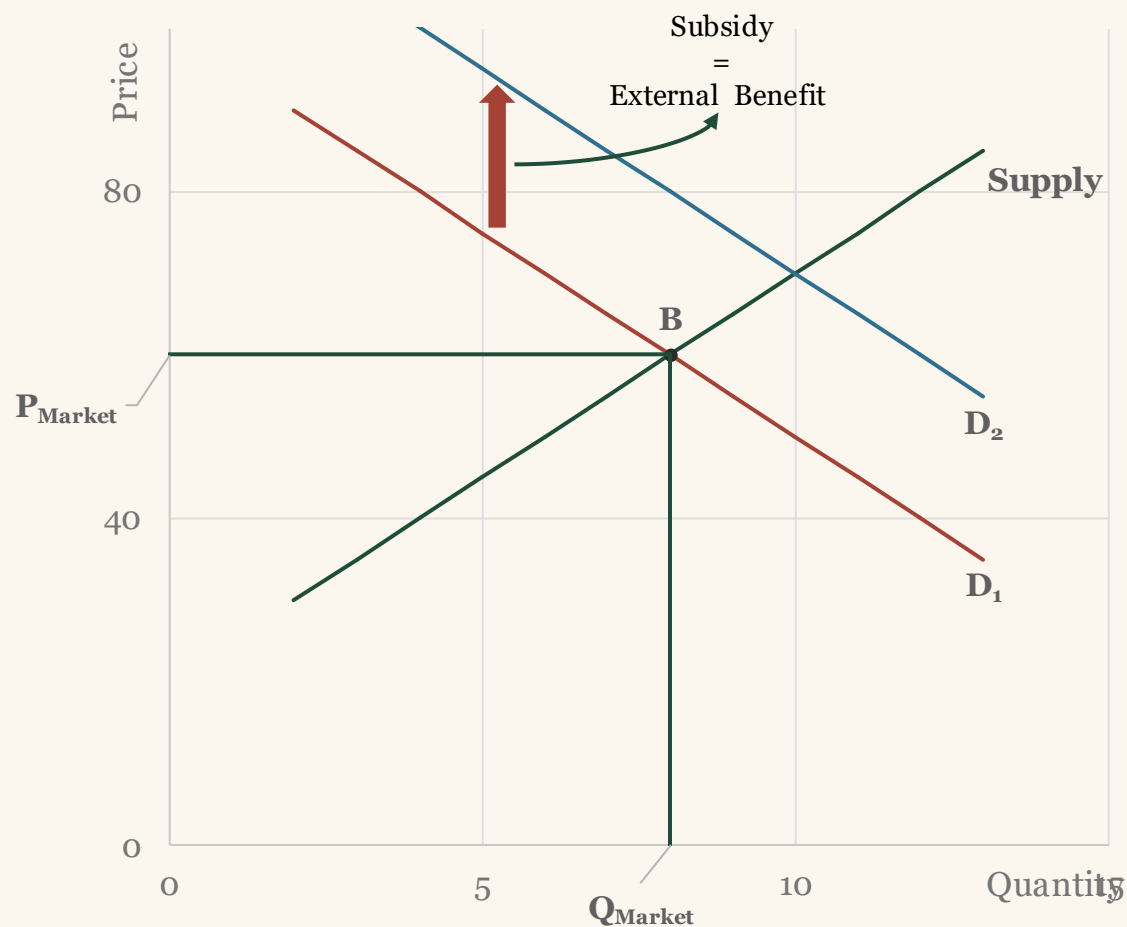
After Tax: $MB = MSC$

$$Q_E = 4$$

Consumers Pay: $P_C = 60$

Bakers Receive: $P_B = 40$

Corrective Subsidies



For a Positive Externality:
Subsidy = External Benefit

Subsidy = 20 Tokens per student

Corrective Subsidies



Subsidy = External Benefit

Subsidy = 20 Tokens per student

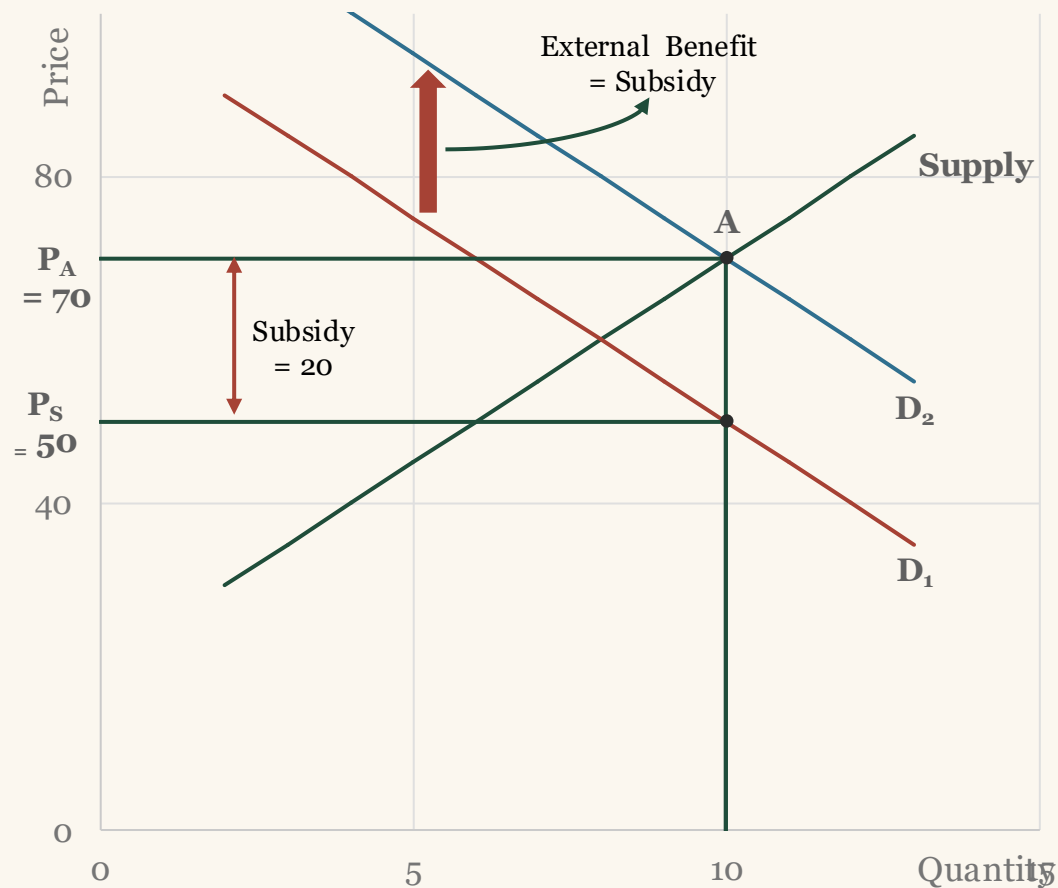
Before Subsidy: $MPB = MC$

$$Q_M = 8$$

After Subsidy: $MSB = MC$

$$Q_E = 10$$

Corrective Subsidies



Subsidy = External Benefit

Subsidy = 20 Tokens per student

After Subsidy: $MSB = MC$

$$Q_E = 10$$

Students Pay: $P_S = 50$

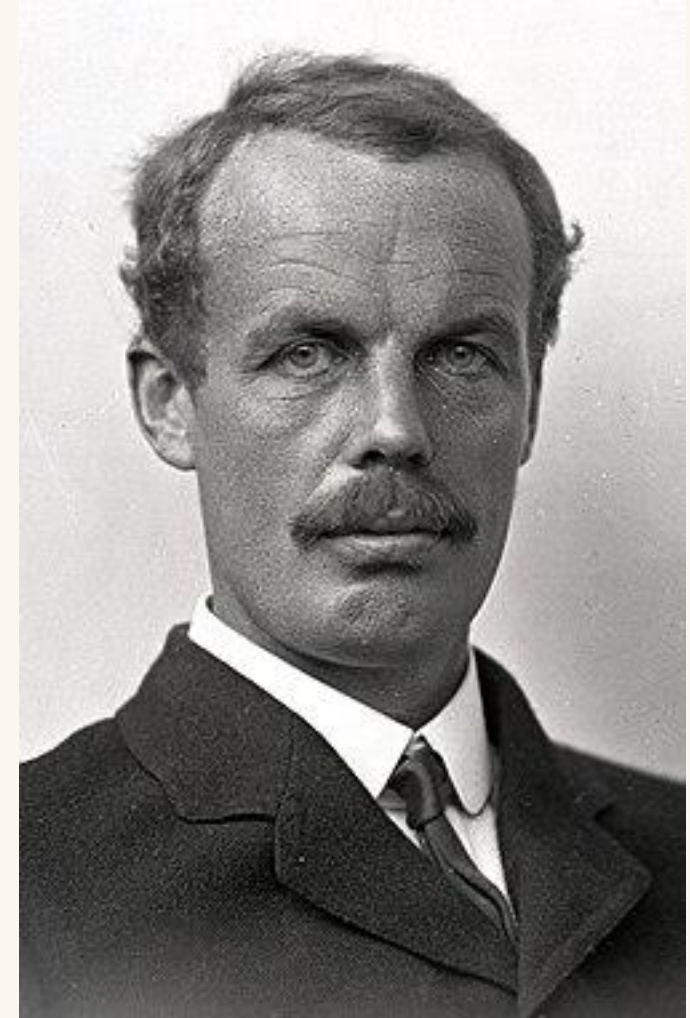
Colleges Receive: $P_A = 70$

Pigouvian Policies

Pigouvian Policies:
Corrective Taxes and Subsidies

Named after **Arthur C. Pigou**

Negative Externality → Tax
Positive Externality → Subsidy



Command and Control Regulation

Government can directly regulate pollution.

Examples:

Maximum soot release, Required chimney filter, Water-quality standard

Problem:

Same rule may impose different costs on different firms.

Tradable Emissions Permits

Bakery	Abatement Cost	Permit Price	Decision
River Café	5	15	Sell permit
Old Mill	25	15	Buy permit

Tradable emissions permits
or

Cap-and-Trade

Government sets total emissions.
Firms trade permits.

Abatement Cost:

Cost of Reducing Pollution

Tradable Emissions Permits

Bakery	Abatement Cost	Permit Price	Decision
River Café	5	15	Sell permit
Old Mill	25	15	Buy permit

River Café gains:

$$15 - 5 = 10$$

Old Mill saves:

$$25 - 15 = 10$$

Same Total Pollution.

Lower Total Cost.